



Disaster-Resilient Communication System

An indigenous solution for critical communication during natural disasters.

SOUTH POINT HIGH SCHOOL



The Challenge: Communication Breakdown

Grid Dependency

Existing systems rely on power and mobile networks, which often collapse.

High Costs

Current infrastructure is expensive and difficult to deploy quickly.

Critical Gap

Communication is lost when it's needed most, hindering relief efforts.

Our project addresses this by leveraging existing street lamp infrastructure for low-cost, off-grid communication.

Scientific Principles & Materials

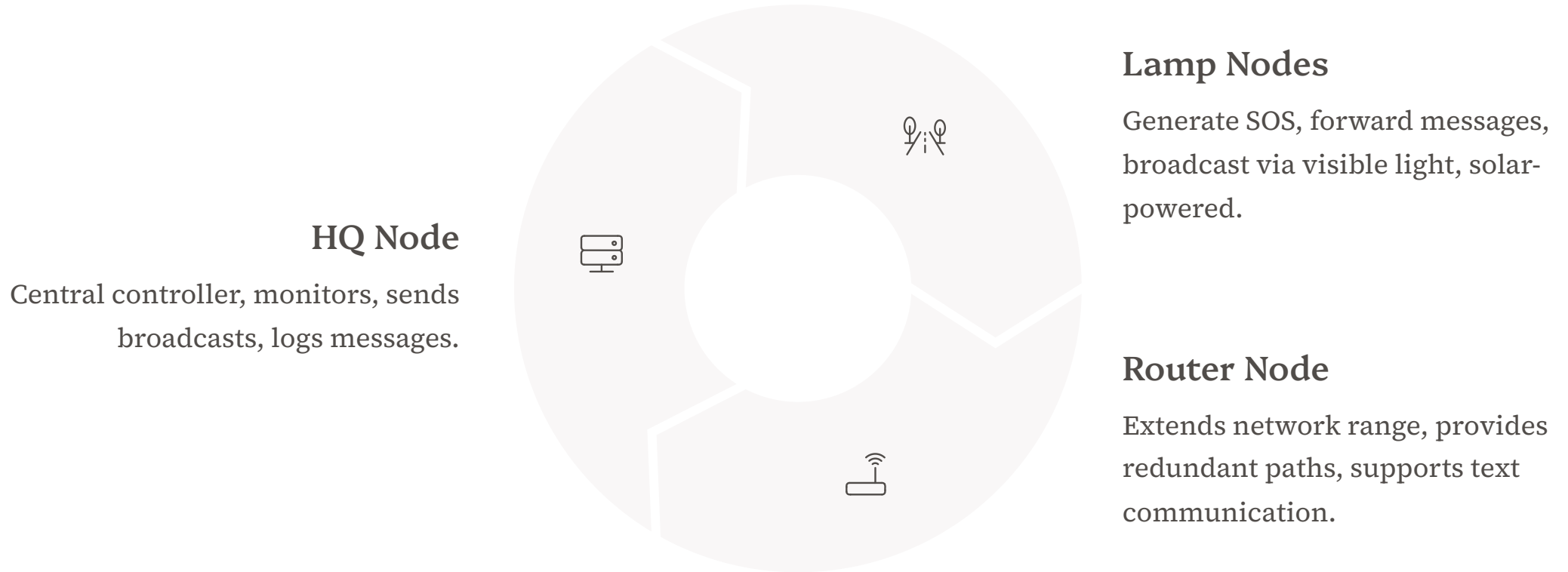
Key Concepts

- Light Fidelity (Li-Fi)
- Ohm's Law
- Light Propagation
- Embedded Systems
- Hashing for Data Integrity

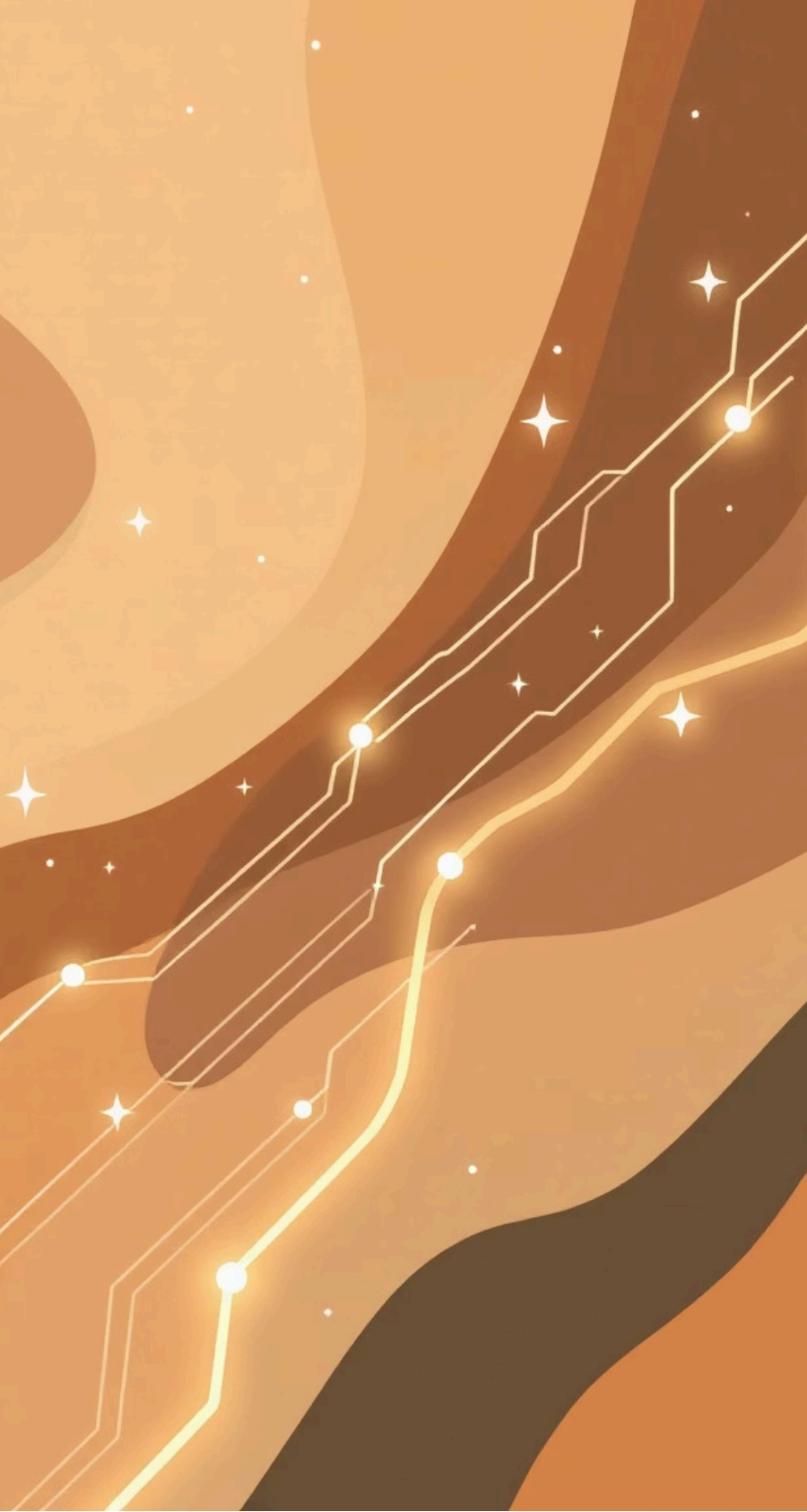
Core Components

- NodeMCU (ESP8266)
- IR Transmitters & Receivers
- LEDs & Resistors
- Batteries & Solar Panels
- SOS Buttons

System Design: Modular Node Roles



This modular design creates a robust, two-way communication mesh for disaster scenarios.



Intelligent Message Routing

1

Uplink (Lamp → HQ)

SOS alerts propagate towards HQ using a hop gradient.

2

Downlink (HQ → Lamps)

HQ broadcasts emergency instructions to all lamps.

1

Deduplication

Cache prevents infinite loops and broadcast storms.

2

IR Retransmission

Multiple transmissions ensure delivery without acknowledgments.

3

Repeating Broadcasts

Lamps retransmit last HQ message via visible light every 60 seconds.

Observations & Future Implications

Key Findings

- Stable transmission up to 20 cm.
- Latency: 1-2 seconds per hop.
- Reliable propagation for 5-6 hops.
- Effective deduplication.
- No significant ambient light interference.

Impact & Scalability

- Works indoors and in low-light.
- Self-healing connectivity.
- Affordable, replicable components.
- Scalable with additional nodes/media.
- Integrates into Smart City infrastructure.